

Financial Econometrics - Homework 7

Chapter 11: Simultaneous Equation Models

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11.24 Reconsider Example 11.2 on the supply and demand for fish at the Fulton Fish Market. The data are in the file *fultonfish*.

- Add the variable *MIXED*, which indicates poor but not *STORMY* weather conditions, to the supply equation in equation (11.14). Estimate the new reduced-form equation for $\ln(\text{PRICE})$, adding the variable *MIXED* to equation (11.16). Is it statistically significant at the 5% level? Test the joint significance of *STORMY* and *MIXED*. Is the resulting *F*-value greater than 10?
- Estimate the demand equation using *STORMY* and *MIXED* as IVs. Compare the coefficient estimates to those in Table 11.5.
- Test the validity of the surplus instrument using the Sargan test, discussed in Section 10.3.4.
- In the reduced-form equation in part (a), test the joint significance of the indicator variables *MON*, *TUE*, *WED*, and *THU* at the 5% level. What do you conclude? Are we now able to estimate the supply equation by 2SLS with confidence in our procedure?

Question 11.24

a. Estimate new reduced-form for $\ln(\text{PRICE})$ with *MIXED* variable.

```
rf_a <- lm(lprice ~ mon + tue + wed + thu + stormy + mixed, data =  
fultonfish)  
summary(rf_a)  
linearHypothesis(rf_a, "mixed = 0")  
linearHypothesis(rf_a, c("stormy = 0", "mixed = 0"))
```

Results:

- *MIXED* variable is statistically significant ($p = 0.0032$).
- Joint significance test for *STORMY* and *MIXED*: $F = 16.14$, $p < 0.0001$.
- Conclusion: Since $F > 10$, the instruments (*STORMY* and *MIXED*) are strong.

b. Estimate demand equation using *STORMY* and *MIXED* as IVs.

```
iv_demand_b <- ivreg(lquan ~ lprice + mon + tue + wed + thu | mon + tue +  
wed + thu + stormy + mixed, data = fultonfish)  
summary(iv_demand_b)
```

Results:

- $\ln(\text{PRICE})$ coefficient: -0.9301 (SE: 0.3534, $p = 0.00977$).

- Comparison: In Table 11.5 (using only STORMY), the price elasticity estimate was -1.119. The new estimate is slightly less elastic but remains negative and significant, consistent with the law of demand.

c. Sargan test for the validity of the surplus instrument.

```
summary(iv_demand_b, diagnostics = TRUE)
```

Results:

- Sargan test statistic: 0.772, p-value = 0.380.

- Conclusion: We fail to reject the null hypothesis. The surplus instrument (MIXED) is valid.

d. Test joint significance of MON, TUE, WED, and THU in reduced form.

```
linearHypothesis(rf_a, c("mon = 0", "tue = 0", "wed = 0", "thu = 0"))
```

Results:

- F-statistic: 0.6279, p-value = 0.6437.

- Conclusion: The indicator variables are not jointly significant in explaining price. This means they are weak instruments for estimating the supply equation. Therefore, we are NOT able to estimate the supply equation by 2SLS with confidence in our procedure.

11.28 Supply and demand curves as traditionally drawn in economics principles classes have price (P) on the vertical axis and quantity (Q) on the horizontal axis.

- a. Rewrite the truffle demand and supply equations in (11.11) and (11.12) with price P on the left-hand side. What are the anticipated signs of the parameters in this rewritten system of equations?
- b. Using the data in the file *truffles*, estimate the supply and demand equations that you have formulated in (a) using two-stage least squares. Are the signs correct? Are the estimated coefficients significantly different from zero?
- c. Estimate the price elasticity of demand “at the means” using the results from (b).
- d. Accurately sketch the supply and demand equations, with P on the vertical axis and Q on the horizontal axis, using the estimates from part (b). For these sketches set the values of the exogenous variables DI , PS , and PF to be $DI^* = 3.5$, $PF^* = 23$, and $PS^* = 22$.
- e. What are the equilibrium values of P and Q obtained in part (d)? Calculate the predicted equilibrium values of P and Q using the estimated reduced-form equations from Table 11.2, using the same values of the exogenous variables. How well do they agree?
- f. Estimate the supply and demand equations that you have formulated in (a) using OLS. Are the signs correct? Are the estimated coefficients significantly different from zero? Compare the results to those in part (b).

Question 11.28

a. Rewrite equations with price P on the left-hand side.

Demand: $P = (-a_1/a_2) + (1/a_2)Q - (a_3/a_2)PS - (a_4/a_2)DI$

Supply: $P = (-b_1/b_2) + (1/b_2)*Q - (b_3/b_2)*PF$

Anticipated Signs:

- Demand: Q coefficient $(1/a_2) < 0$; PS coefficient $(-a_3/a_2) > 0$; DI coefficient $(-a_4/a_2) > 0$.
- Supply: Q coefficient $(1/b_2) > 0$; PF coefficient $(-b_3/b_2) > 0$.

b. 2SLS estimation of rewritten equations.

```
iv_demand_p <- ivreg(p ~ q + ps + di | ps + di + pf, data = truffles)
iv_supply_p  <- ivreg(p ~ q + pf | ps + di + pf, data = truffles)
summary(iv_demand_p)
summary(iv_supply_p)
```

Results:

- Demand: Q: -2.671 (p=0.031), PS: 3.461 (p=0.004), DI: 13.390 (p<0.001). All signs match expectations and coefficients are significant.
- Supply: Q: 2.936 (p<0.001), PF: 2.958 (p<0.001). All signs match expectations and coefficients are significant.

c. Price elasticity of demand at the means.

```
g2 <- coef(iv_demand_p) ["q"]
elasticity <- (1/g2) * (mean(truffles$p) / mean(truffles$q))
```

Results:

- Mean P: 62.72, Mean Q: 18.46.
- Calculated Elasticity: -1.272.

d. Sketch intercepts and slopes (DI=3.5, PF=23, PS=22).

From R results:

- Demand: $P = 111.58 - 2.67 * Q$
- Supply: $P = 9.25 + 2.94 * Q$

e. Equilibrium values and reduced-form comparison.

```
# Calculated from equations
Q_eq: 18.25, P_eq: 62.84
# Predicted from Table 11.2 reduced form
Q_pred: 18.26, P_pred: 62.82
```

Conclusion: The values agree very well.

f. OLS estimation comparison.

```
ols_demand_p <- lm(p ~ q + ps + di, data = truffles)
summary(ols_demand_p)
```

Results:

- Demand: Q coefficient in OLS is 0.151 ($p=0.764$), which is positive (incorrect sign) and insignificant.
- Conclusion: OLS fails to estimate the demand curve correctly due to simultaneity bias, whereas 2SLS provides a consistent negative and significant slope.